A Three-Sector Formalism for Energy–Mass–Information with Contextual Conversion, ABS-Stable Readout, and Admissibility Gates

Abstract

We formalize a common intuition appearing in operator-based simulation pipelines: *energy, mass, and information are mutually related but not mutually* ***closed*** *under a single, context-free reduction.* We provide a minimal mathematical framework in which (i) internal dynamics are norm-preserving (unitary core), (ii) observable signals are obtained by an ABS-stable readout (robust near phase singularities), and (iii) a *Gate* enforces admissibility (stability, auditability, and optionally hard constraints). The main theorems are *no-go results*: there is no universal, parameter-free bijection converting energy+mass into information (or any two sectors into the third) while preserving the natural additive invariants of each sector. Any conversion necessarily depends on additional context (e.g., encoding map, temperature, reference frame). This reconciles "exponentially growing readout signals" with conservation: growth may occur in an open/controlled channel or in a nonlinear readout, while the conservative core remains norm-preserving.

1 Executive summary (dry)

- **Core:** internal state $\psi(t) \in \mathcal{H}$ evolves conservatively (unitary or otherwise norm-bounded).
- **Readout:** a scalar observable is extracted by an ABS map $A(t) = |M_t(\psi(t))|$, stable at nodes.
- **Gate:** an admissibility operator G restricts what is allowed to be measured/acted upon.
- **Main point:** *Energy, mass, information* are related by *contextual* bridges (c^2, $k_B T \ln 2$, encoding/decoding), not by a single context-free identification.

2 Three sectors: what is meant by "energy = mass = information"

2.1 Minimal state model

Let (Ω, μ) be a measurable space and let $\mathcal{H}$ be a complex Hilbert space of latent states (e.g. $\mathcal{H} = L^2(\Omega, \mu)$ or $\mathcal{H} = \mathbb{C}^d$). We treat a *physical realization* as a pair $(\psi, \mathcal{E})$ where $\psi \in \mathcal{H}$ is a state and $\mathcal{E}$ is a choice of *experiment/encoding context*.

Definition 1 (Energy functional)**.** *An* energy functional *is a map $E : \mathcal{D}(E) \subset \mathcal{H} \times \mathcal{E} \to \mathbb{R}_{\geq 0}$ such that for fixed context $\mathcal{E}$ it is additive under independent composition and is bounded below.*

Definition 2 (Mass functional)**.** *A* mass functional *is a map $M : \mathcal{D}(M) \subset \mathcal{H} \times \mathcal{E} \to \mathbb{R}_{\geq 0}$ that assigns an inertial/rest mass parameter to the realized system under a chosen reference frame (again part of $\mathcal{E}$).*

Definition 3 (Information functional)**.** *Fix an* encoding $\mathrm{Enc}_{\mathcal{E}} : \mathcal{H} \to \mathcal{Y}$ *into an alphabet/model class $\mathcal{Y}$, a probability model $P_{\mathcal{E}}$ on $\mathcal{Y}$, and a decoding/observer convention. An* information functional *is a map $I : \mathcal{D}(I) \subset \mathcal{H} \times \mathcal{E} \to \mathbb{R}_{\geq 0}$ defined by an information measure on $\mathcal{Y}$ (e.g. Shannon entropy, Kolmogorov description length, or a task-specific code length).*

Remark 1 (Why this is not pedantry). Information is not a scalar attached to a bare ψ; it depends on a *codebook* and an *observer model.* That dependence is the technical core of the "cannot close everything into one sector" claim.

2.2 Contextual bridges (not identifications)

There exist *bridges*:

- $E \leftrightarrow M$ is bridged by $E = c^2 M$ *after* specifying a frame and what portion is rest energy vs. kinetic/binding energy (context).
- $E \leftrightarrow I$ is bridged by thermodynamic constraints, e.g. a lower bound of the form $E \gtrsim k_B T \ln 2 \cdot (\text{bits erased})$, which depends on temperature T (context).
- $M \leftrightarrow I$ is bridged only via a physical substrate and an encoding/decoding convention; both are context.

The formal results below show that without such context, no universal conversion can be additive and invertible.

3 No-go theorems: why two sectors do not close the third without context

3.1 Additive invariants

We model each sector as an additive commutative monoid:

$$(\mathsf{E}, +), \qquad (\mathsf{M}, +), \qquad (\mathsf{I}, +),$$

where "+" denotes composition of independent subsystems or concatenation of independent messages under a fixed context. This captures the operational invariance "two independent things add their totals".

Theorem 1 (No context-free closure to information)**.** *There is no map*

$$F : \mathsf{E} \times \mathsf{M} \to \mathsf{I}$$

that is simultaneously:

1. ***Additive** in each argument: $F(E_1 + E_2, M) = F(E_1, M) + F(E_2, M)$ and $F(E, M_1 + M_2) = F(E, M_1) + F(E, M_2)$,*
2. ***Injective** (distinct (E, M) yield distinct information),*
3. ***Context-free** (does not depend on any additional parameter such as a codebook, temperature, or frame).*

Proof. Assume such an F exists. Fix any M_0. Additivity implies $F(E, M_0) = E \cdot F(1, M_0)$ in the sense of repeated addition (monoid homomorphism), so $F(\cdot, M_0)$ is completely determined by a single constant $c(M_0) := F(1, M_0)$. Similarly, fixing E_0 yields $F(E_0, M) = M \cdot d(E_0)$. Consistency for general (E, M) forces $F(E, M) = E\, c(M) = M\, d(E)$, which implies a rigid separable form. Now pick $M_0 \neq M_1$ with $c(M_0) \neq c(M_1)$ (otherwise F ignores M and cannot be injective). Then for any E, $F(E, M_0) - F(E, M_1) = E\,(c(M_0) - c(M_1))$ can be made arbitrarily large or small by varying E, forcing a linear scale choice that effectively *selects a unit* and *an encoding* for I. But information units are defined only relative to an encoding/model class; without context, there is no distinguished scale. Concretely: choose two different encodings (two different I-monoid coordinatizations) that rescale "one bit" by a constant factor; then F cannot be simultaneously additive and context-free because it would have to pick one scale. This contradicts context-freedom. Hence such F cannot exist. □

Remark 2. The proof isolates the obstruction: **information requires a codebook/observer convention**. Any numeric information value is defined only up to a context-dependent scaling and reference.

Theorem 2 (No context-free closure to energy). *There is no map $G : \mathsf{I} \times \mathsf{M} \to \mathsf{E}$ that is additive, injective, and context-free.*

Proof. Energy has physical dimensions and depends on state of motion/binding; mass-energy conversion requires a reference frame and specification of what is counted as rest vs. kinetic/binding energy. Information-to-energy conversion is not defined without a thermodynamic context (temperature, entropy budget, and an explicit physical erasure/measurement process). Without such parameters, any additive assignment $G(I, M)$ would contradict invariance under change of units/frames: one may change I by changing encoding while leaving the physical system unchanged, which would force energy to change despite identical physics. Thus a context-free injective G cannot exist. □

Corollary 1 ("Related but not identical"). *Statements like "energy = mass = information" should be read as:* the three sectors are coupled by contextual bridges *(constants and protocols), not as* an unconditional identity between scalars.

4 Unitary core + ABS-first readout: how exponential-looking signals arise without free energy

4.1 Conservative internal evolution

Let $U(t)$ be a unitary evolution on $\mathcal{H}$:

$$U(t)^* U(t) = I, \qquad \psi(t) = U(t)\psi(0).$$

Then $\|\psi(t)\| = \|\psi(0)\|$ for all t.

4.2 ABS-stable readout (branch-safe)

Fix a bounded linear functional $M_t : \mathcal{H} \to \mathbb{C}$ (a probe/measurement). Define the observable

$$A(t) := |M_t(\psi(t))| .$$

This map $\psi \mapsto |M_t(\psi)|$ is *nonlinear* and can display sharp growth/contrast due to interference and changing probes, even when $\|\psi(t)\|$ is constant.

Lemma 1 (Phase safety). *For any global phase $\alpha \in \mathbb{R}$, $A(t)$ is invariant under $\psi(t) \mapsto e^{i\alpha}\psi(t)$. Moreover, $A(t)$ remains well-conditioned near nodes where $\arg(M_t(\psi(t)))$ is ill-defined.*

Proof. $|M_t(e^{i\alpha}\psi)| = |e^{i\alpha} M_t(\psi)| = |M_t(\psi)|$. Nodes correspond to $M_t(\psi) = 0$ where argument is undefined; magnitude is continuous and robust. □

4.3 Chaos and harmony coexistence as orthogonal components

Let $\psi(t) = \psi_h(t) + \psi_c(t)$ with $\psi_h \perp \psi_c$. Define normalized energy fractions (in the Hilbert norm sense):

$$\eta_h(t) := \frac{\|\psi_h(t)\|^2}{\|\psi(t)\|^2}, \qquad \eta_c(t) := \frac{\|\psi_c(t)\|^2}{\|\psi(t)\|^2}, \qquad \eta_h(t) + \eta_c(t) = 1.$$

Both components can be nonzero simultaneously; a projection/readout can emphasize one and suppress the other.

Proposition 1 (Coexistence under unitary mixing). *If $\psi(t) = U(t)\psi(0)$ is unitary and ψ_h, ψ_c are defined by fixed orthogonal projectors P_h, P_c with $P_h + P_c = I$, then $\eta_h(t), \eta_c(t)$ vary with t but remain in $[0, 1]$ and sum to 1.*

Proof. $\psi_h = P_h\psi$, $\psi_c = P_c\psi$. Orthogonality gives $\|\psi\|^2 = \|P_h\psi\|^2 + \|P_c\psi\|^2$. Unitary evolution preserves $\|\psi\|$, hence the stated constraints. □

5 The Gate: admissibility, stability, and (optional) hard constraints

5.1 Gate as a measurable envelope

Let $G : \mathcal{H} \times \mathcal{E} \to \mathcal{H} \times \mathcal{E}$ be an admissibility operator that enforces stability and auditability (e.g. bounded gradients, tail control, or constraint satisfaction). The operational pipeline is:

$$\psi(t) \xrightarrow{\text{core}} \psi(t) \xrightarrow{G} \psi_G(t) \xrightarrow{\text{readout}} A(t) = |M(\psi_G(t))|.$$

5.2 A hard absorbing boundary (optional "moral firewall" formalism)

Let $\mathcal{A}$ be a set of actions and let $\mathcal{R} \subset \mathcal{A}$ be forbidden actions. Define a hard filter

$$\Pi_{\text{safe}}(a) = \begin{cases} a, & a \notin \mathcal{R}, \\ \perp, & a \in \mathcal{R}. \end{cases}$$

This is an admissibility cut (domain restriction), not an energy term. It can be coupled to SAT/UNSAT witness reporting in discrete implementations.

6 Engineer appendix: numerical stability and "sulin_div"

6.1 Why finite precision breaks ideal algebra

In floating point arithmetic, denominators that are theoretically zero often appear as small nonzero numbers, causing spurious blow-ups. A common stabilization is an ε-guarded division.

Definition 4 (`sulin_div` pattern)**.** *Given $a, b \in \mathbb{R}$ and $\varepsilon \geq 0$, define*

$$\text{sulin\_div}(a, b; \varepsilon) := \begin{cases} a, & |b| \leq \varepsilon, \\ a/b, & |b| > \varepsilon. \end{cases}$$

Remark 3. This is a *numerical Gate*: it does not change the mathematical model, it prevents arithmetic singularities from dominating the computation. It is consistent with the philosophy "stability before interpretation".

7 Conclusion

Energy, mass, and information are coupled but not collapsible into a single sector without specifying the missing context (encoding, temperature, frame, protocol). In a conservative (unitary) core, large apparent signal growth can occur in an ABS-stable nonlinear readout or via an explicit open/controlled channel, without implying free energy. The practical implementation principle is: *define the Gate, prove stability, then trust the readout.*